

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 81. Card Payment Receipt - Project

Section 1: Lecture Summary

This lecture covers a project about formatting credit or debit card numbers in transaction alerts, focusing on privacy by masking the card number except for the last four digits. The scenario simulates receiving an SMS message after a card transaction, where the card number should be displayed with the first 12 digits replaced by 'X' and only the last four digits revealed. The input is a 16-digit card number with spaces included, and the output formats this number accordingly. The instructor explains how to slice strings in Python to extract the last four digits and create a masked string for the first 12 digits using repeated 'X' characters and spaces. Finally, the lecture shows how to combine these parts and print a formatted string that resembles a real transaction alert message.

Section 2: Key Concepts and Explanations

- Credit Card Number Format

- A typical credit card number consists of 16 digits.
- It is often displayed with spaces every 4 digits (total string length 19 if spaces are included).
- For privacy, only last four digits are shown openly, the rest are masked.

- String Slicing in Python

- Strings can be sliced using ``string[start:end]`` notation.
- To extract the last four digits from a 19-character input (including spaces), slice from index 15 onward: ``card_number[15:]``.

- Masking the Card Number

- The first 12 digits are replaced by 'X's. Since the digits are grouped in 4s separated by spaces (e.g., `XXXX XXXX XXXX`), the masking involves creating three groups of 'XXXX ' (four X characters plus a space).

- Achieved by:

```
four = 'X' * 4 + ' ' # group of 4 X's plus space
masked_part = four * 3 # repeat 3 times to cover 12 digits
```

- Concatenating the Masked and Visible Parts

- Combine the masked groups with the last four digits to create the final display string:

```
`display_number = masked_part + last_digits`
```

- Full Transaction Alert Message

- The final output message can include the transaction alert text with the masked card number embedded.

Section 3: Example Code and Use Cases

Take card number input including spaces

Extract last four digits starting from index 15 (assuming format with spaces)

Create one group of 4 'X's followed by a space

Repeat the masked group 3 times to cover first 12 digits

Combine masked part with last four digits

Print the masked card number

```
card_number = input("Enter Card No: ") # e.g., "4455 1122 3344 5566"

last_digits = card_number[15:]

four = 'X' * 4 + ' '

masked_part = four * 3
```

```
display_number = masked_part + last_digits  
print(display_number)
```

Explanation:

- Input expects a card number with spaces, length 19 characters.
- The last 4 digits are sliced from index 15 onwards.
- The first 12 digits are replaced by 3 groups of 'XXXX '.
- The final masked card number resembles: 'XXXX XXXX XXXX 5566'.

Optional full message printout:

```
amount = 1000 # example amount debited  
print(f"Transaction Alert\nAmount ${amount} has been debited from your  
card\n{display_number}.")
```

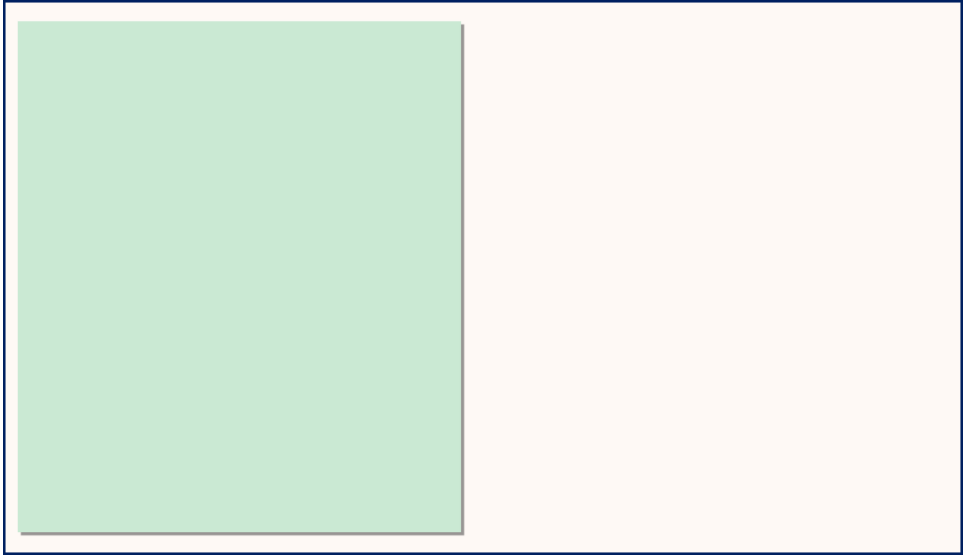
Section 4: Key Takeaways

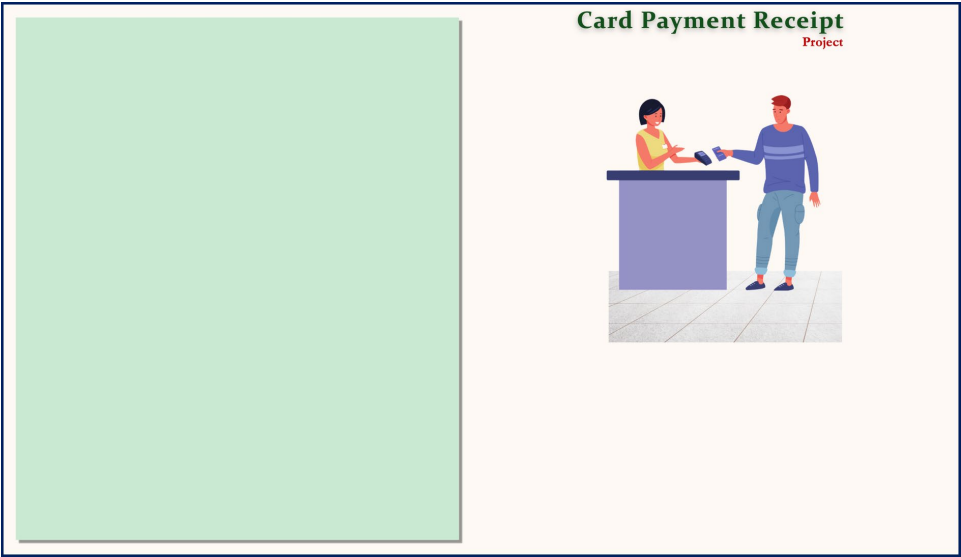
- Credit card numbers should be masked for privacy, showing only last four digits.
- Python string slicing is a powerful and simple tool to extract specific portions of a string.
- String repetition (`` operator) can efficiently create repeated patterns like 'XXXX '.
- Combining string slices and repeated strings enables formatting structured outputs such as masked card numbers.
- This project simulates a practical real-world scenario of formatting transaction alert messages securely.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.

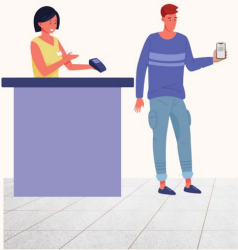


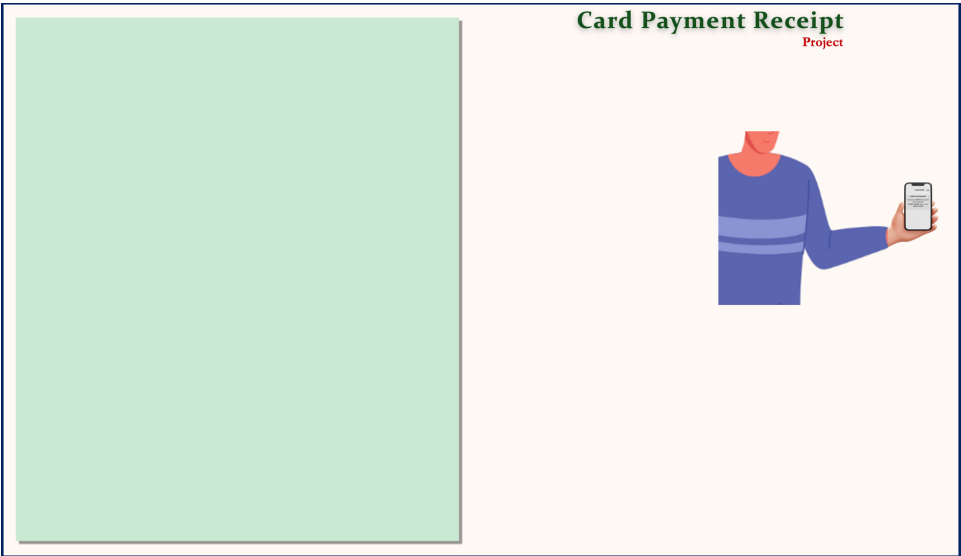
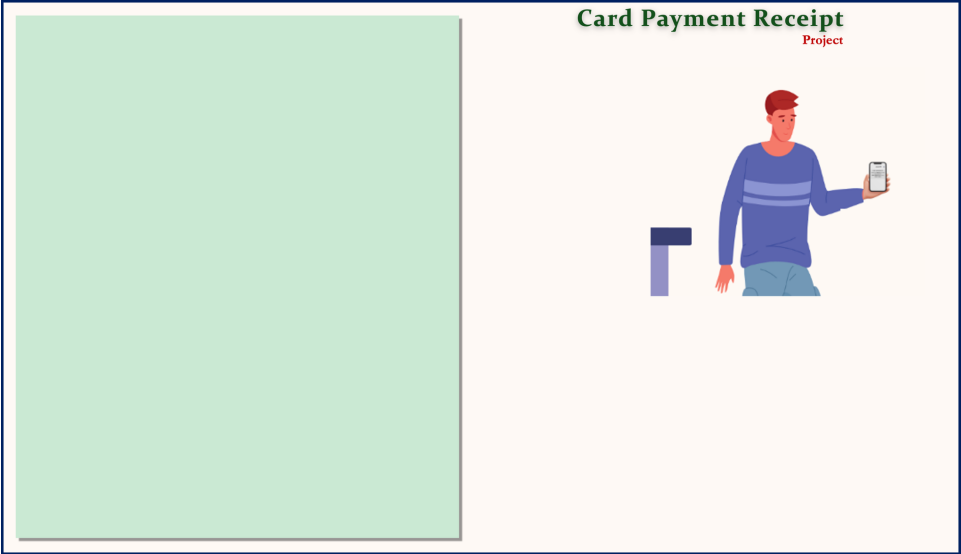


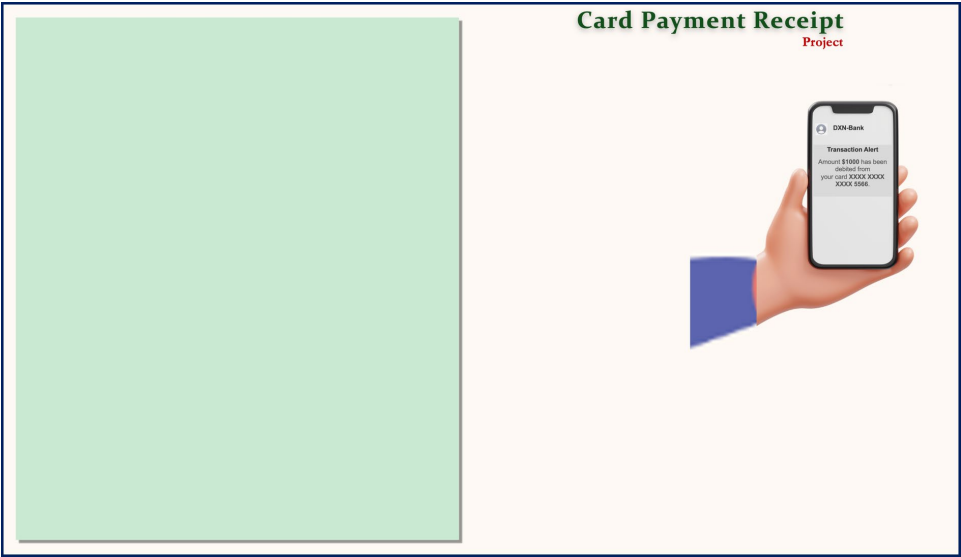
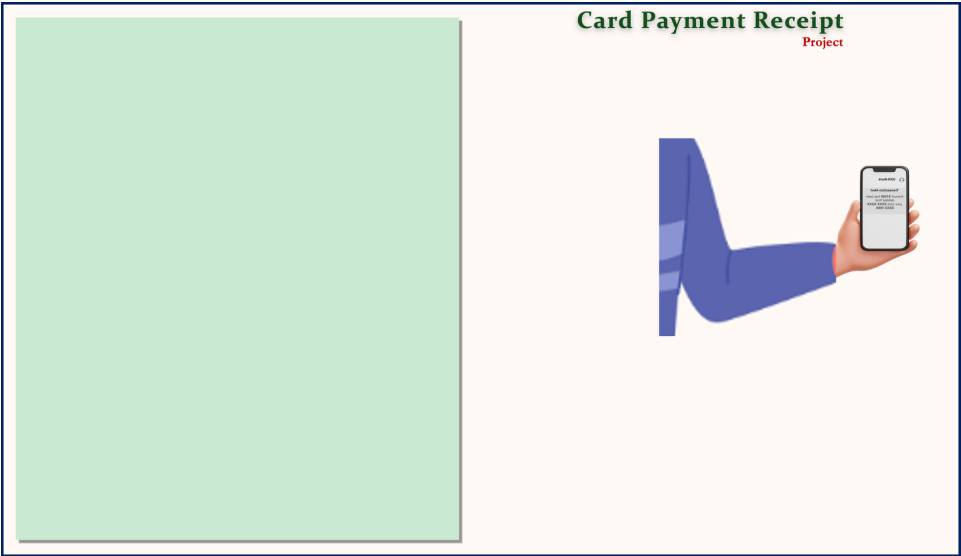


Card Payment Receipt

Project

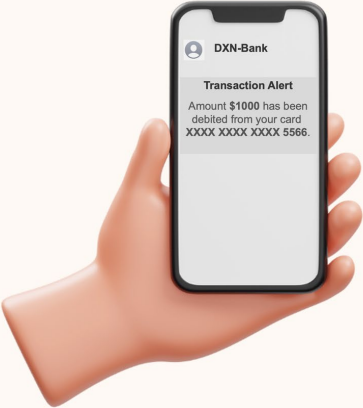
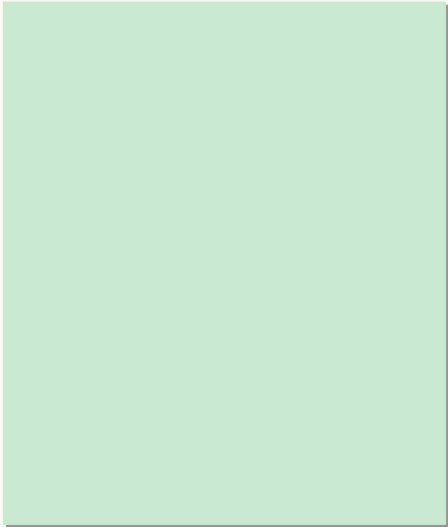






Card Payment Receipt

Project



Card Payment Receipt

Project

Transaction Alert

Amount \$1000 has been debited from your card
XXXX XXXX XXXX 5566.

Card Payment Receipt

Project

Transaction Alert

Amount \$1000 has been debited from your card
XXXX XXXX XXXX 5566.

Card No 4455 1122 3344 5566

Card Payment Receipt

Project

Transaction Alert

Amount \$1000 has been debited from your card
XXXX XXXX XXXX 5566.

Card No4455 1122 3344 5566

Disp NoXXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo4455 1122 3344 5566

DispNoXXXX XXXX XXXX 5566

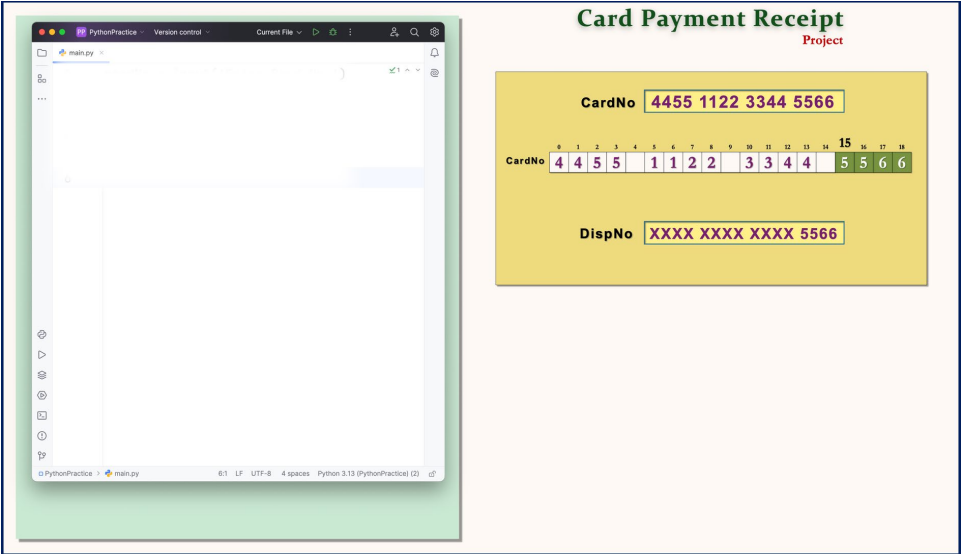
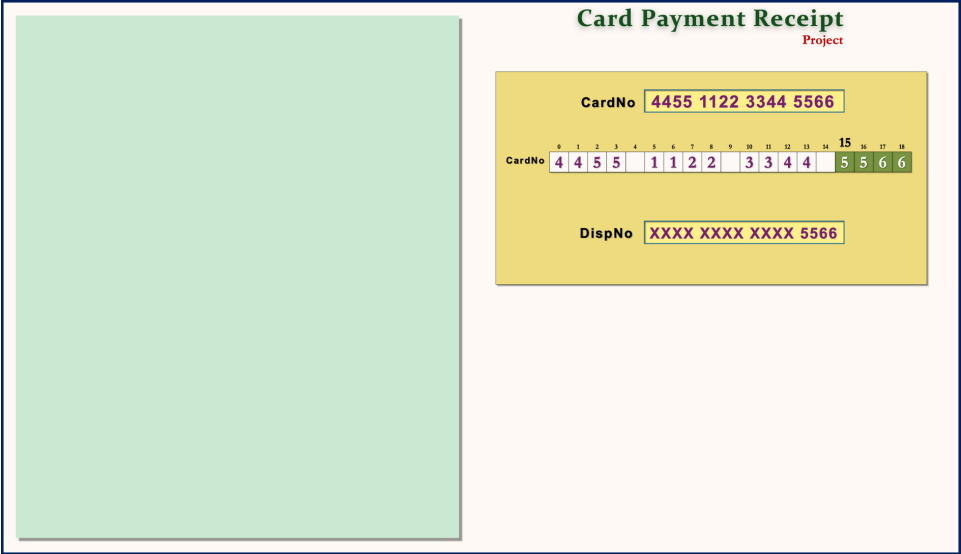
Card Payment Receipt

Project

CardNo 4455 1122 3344 5566

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
CardNo	4	4	5	5		1	1	2	2		3	3	4	4		5	5	6	6

DispNo XXXX XXXX XXXX 5566



PythonPractice

main.py

...

Run

main

/Users/abdulbari/PycharmProjects/PythonPracti

Enter Card No|

PythonPractice

main.py

6:1

LF

UTF-8

4 spaces

Python 3.13 (PythonPractical (2))

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

4455112233445566

DispNo

XXXX XXXX XXXX 5566

PythonPractice

main.py

...

Run

main

/Users/abdulbari/PycharmProjects/PythonPracti

Enter Card No 4546 1234 5678 5877

XXXX XXXX XXXX 5877

Process finished with exit code 0

PythonPractice

main.py

6:1

LF

UTF-8

4 spaces

Python 3.13 (PythonPractical (2))

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

4455112233445566

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

4455112233445566

lastdigits =

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0123456789101112131415161718

4455112233445566

lastdigits = CardNo[]

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0123456789101112131415161718

4455112233445566

lastdigits = CardNo[15]

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four =

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four = 'X' * 4

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo 4455 1122 3344 5566

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
CardNo	4	4	5	5		1	1	2	2		3	3	4	4		5	5	6	6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo =

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo

4455 1122 3344 5566

CardNo

0

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3

DispNo

XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo4455112233445566

CardNo

8

7

6

5

4

3

2

1

0

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3 + lastdigits

DispNoXXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo4455112233445566

CardNo

8

7

6

5

4

3

2

1

0

15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

4

4

5

5

1

1

2

2

3

3

4

4

5

5

6

6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3 + lastdigits

DispNoXXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo 4455 1122 3344 5566

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
CardNo	4	4	5	5		1	1	2	2		3	3	4	4		5	5	6	6

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3 + lastdigits

print(DispNo)

DispNo XXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo4455112233445566

0123456789101112131415161718

CardNo4455112233445566

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3 + lastdigits

print(DispNo)

DispNoXXXX XXXX XXXX 5566

Card Payment Receipt

Project

CardNo4455112233445566

0123456789101112131415161718

CardNo4455112233445566

lastdigits = CardNo[15 :]

four = 'X' * 4 + ' '

DispNo = four * 3 + lastdigits

print(DispNo)

DispNoXXXX XXXX XXXX 5566